ASSIGNMENT 4.2

Knowledge Representation and Adaptive API Integration

Student	[Student Name]	Roll No	[Roll No]
Course	21CSE525T - Agent Technology	Project	Placement Preparation Agent

Unit 4 - Ontology with inference, plus live REST data that changes the plan. File: assignments/a4_2/knowledge_api_agent.py

1. Objective

Represent placement knowledge as a graph, derive new facts by inference, and let live web data change the agent's plan.

2. Knowledge graph

The ontology has skills, companies and resources as individuals, linked by five typed relations: REQUIRED_BY (skill to company), PREREQUISITE_OF (skill to skill), TEACHES (resource to skill), HAS_LEVEL_IN (student to skill) and ELIGIBLE_FOR (student to company). It is stored as JSON.

Inference 1 - prerequisite closure: prerequisites are followed transitively. For example, dp needs arrays and recursion, and system_design needs dbms and oops.

Inference 2 - skill gap: the difference between the student's levels and a company's requirements.

Inference 3 - eligibility: companies whose CGPA and skill requirements the student meets.

3. Live data

- GitHub Users API: public repository count and followers, used as a portfolio signal.
- Open-Meteo: current temperature and precipitation, used to choose a study modality.
- Each call has a timeout and a fallback. If GitHub is rate-limited or unreachable, the agent records the failure and uses a last-known-good payload, so the demo always runs. The offline flag forces this path.

4. Adaptation rules

- Rule 1: if public repositories are 5 or more, the portfolio task is suppressed and its hours go to the weakest skill.
- Rule 2: if rainfall is above 0.5 mm, all tasks switch to indoor-online and offline travel is cancelled. Otherwise, if temperature is above 33 C, all tasks move to an early-morning slot (05:30 to 08:30). Otherwise the normal schedule stays.

5. Implementation

```
class KnowledgeGraph:
    def __init__(self, kb):
        self.kb = kb

    def save(self, path=KB_FILE):
        with open(path, "w") as f:
            json.dump(self.kb, f, indent=2)
        log("KR", f"ontology serialised -&gt; {path} ({os.path.getsize(path)} bytes)")

    # ---- inference: transitive prerequisite closure -------------------
    def prerequisite_closure(self, skill, seen=None):
        seen = seen or set()
        for p in self.kb["individuals"]["skills"].get(skill, {}).get("prerequisites", []):
            if p not in seen:
                seen.add(p)
                self.prerequisite_closure(p, seen)
        return seen

    # ---- inference: skill gap query ----------------------------------
```

```
    def skill_gap(self, student_levels, company):
        req = self.kb["individuals"]["companies"][company]["required_skills"]
        gaps = []
        for s in req:
            lvl = student_levels.get(s, 0)
            if lvl &lt; 70:
                gaps.append((s, lvl, sorted(self.prerequisite_closure(s))))
        return gaps

    def eligible_companies(self, cgpa, levels):
        out = []
        for name, c in self.kb["individuals"]["companies"].items():
            if cgpa &gt;= c["cgpa_cutoff"]:
                cov = sum(1 for s in c["required_skills"] if levels.get(s, 0) &gt;= 70)
                out.append((name, c["tier"], f"{cov}/{len(c['required_skills'])}"))
        return sorted(out, key=lambda x: x[1])

    def resources_for(self, skill):
        return [r for r, v in self.kb["individuals"]["resources"].items()
                if skill in v["teaches"]]

def fetch_github_profile(user="torvalds"):
    """REAL REST API: https://api.github.com/users/&lt;user&gt;"""
    url = f"https://api.github.com/users/{user}"
    if OFFLINE:
        log("API/GITHUB", "offline mode -&gt; using cached mock payload")
        return MOCK_GITHUB, "MOCK"
    # An authenticated request raises the quota from 60/hr per IP to 5000/hr, which
    # matters on a shared campus NAT where the anonymous quota is usually spent.
    headers = {}
    token = os.environ.get("GITHUB_TOKEN", "").strip()
    if token:
        headers["Authorization"] = f"Bearer {token}"
    try:
        log("API/GITHUB", f"GET {url}" + ("  [authenticated]" if token else "  [anonymous]"))
        data = http_get_json(url, extra_headers=headers)
        log("API/GITHUB", f"200 OK -&gt; repos={data.get('public_repos')} "
                          f"followers={data.get('followers')}")
        return data, "LIVE"
    except urllib.error.HTTPError as e:
        # 403 + rate-limit is an EXPECTED operating condition, not a defect. Naming it
        # precisely is the difference between "the demo broke" and "degradation worked".
        if e.code in (403, 429):
            log("API/GITHUB", "403 rate-limited (anonymous quota is 60/hr per IP) "
                              "-&gt; resilience path: serving last-known-good payload")
            return MOCK_GITHUB, "CACHED-RATELIMIT"
        log("API/GITHUB", f"HTTP {e.code} -&gt; resilience path: last-known-good payload")
        return MOCK_GITHUB, "CACHED-HTTP%d" % e.code
    except Exception as e:
        log("API/GITHUB", f"unreachable ({type(e).__name__}) "
                          f"-&gt; resilience path: last-known-good payload")
        return MOCK_GITHUB, "CACHED-OFFLINE"
```

6. Result

In the recorded run GitHub returned live data (12 public repositories), and Open-Meteo returned 31.5 C with no rain. Rule 1 suppressed the portfolio task because the repository count reached 5. The plan changed from live data: 2 adaptive mutations on a 4-task plan.

Execution output (log file)

Captured from **logs/4.2_knowledge_api_agent.log** (python3 knowledge_api_agent.py). Timestamps and some values differ between runs; this is the recorded run.

```
# Activity 4.2_knowledge_api_agent
# command: python3 knowledge_api_agent.py
# exit code: 0


================================================================================================
 KNOWLEDGE REPRESENTATION + ADAPTIVE API INTEGRATION (Assignment 4.2)
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 Mode: LIVE (real REST endpoints)

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 PART 1 :: ONTOLOGICAL KNOWLEDGE REPRESENTATION
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```

```
   Ontology v1.2 | 4 classes | 5 relation types
      (Skill) --[REQUIRED_BY]--&gt; (Company)
      (Skill) --[PREREQUISITE_OF]--&gt; (Skill)
      (Resource) --[TEACHES]--&gt; (Skill)
      (Student) --[HAS_LEVEL_IN]--&gt; (Skill)
      (Student) --[ELIGIBLE_FOR]--&gt; (Company)
   Individuals: 11 skills, 5 companies, 5 resources
[12:26:32] [KR            ] ontology serialised -&gt; knowledge_base.json (4074 bytes)

   INFERENCE QUERIES OVER THE GRAPH:
      prerequisite_closure('dp')           = ['arrays', 'recursion']
      prerequisite_closure('system_design') = ['dbms', 'oops']
      resources_for('graphs')              = ['striver_a2z', 'neetcode_150']

      eligible_companies(cgpa=7.6):
         Zoho         tier-2  skill coverage 3/4
         Freshworks   tier-2  skill coverage 3/4
         Infosys      tier-3  skill coverage 3/3
         TCS          tier-3  skill coverage 2/3

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 PART 2 :: BASE PLAN DERIVED PURELY FROM THE KNOWLEDGE GRAPH
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    BASE PLAN (target = Amazon)
    #  Skill             Lvl  Hrs  Mode               Blocked-by           Resources
    1  system_design      22   12  deep-study         -                    ddia_book
    2  dp                 38   16  deep-study         ['recursion']        striver_a2z,neetcode_150
    3  graphs             41   16  deep-study         ['recursion']        striver_a2z,neetcode_150
    4  trees              66   16  deep-study         ['recursion']        striver_a2z

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 PART 3 :: LIVE REST API INGESTION
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[12:26:32] [API/GITHUB    ] GET https://api.github.com/users/torvalds  [anonymous]
[12:26:33] [API/GITHUB    ] 200 OK -&gt; repos=12 followers=329594
[12:26:33] [API/WEATHER   ] GET https://api.open-meteo.com/v1/forecast?latitude=12.92&amp;longitude=80.12&amp;...
[12:26:33] [API/WEATHER   ] 200 OK -&gt; 31.5C humidity=49% rain=0.0mm

   github  source=LIVE  public_repos=12 followers=329594
   weather source=LIVE  temp=31.5C precip=0.0mm

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 PART 4 :: ADAPTIVE BEHAVIOURAL EXECUTION (plan mutates from live data)
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[12:26:33] [ADAPT         ] 1. public_repos=12 &gt;= 5 -&gt; portfolio task SUPPRESSED, hours reallocated to weakest skill
[12:26:33] [ADAPT         ] 2. temperature=31.5C, rain=0.0mm -&gt; normal schedule retained

    ADAPTED PLAN (post-ingestion)
    #  Skill             Lvl  Hrs  Mode               Blocked-by           Resources
    1  system_design      22   18  deep-study         -                    ddia_book
    2  dp                 38   16  deep-study         ['recursion']        striver_a2z,neetcode_150
    3  graphs             41   16  deep-study         ['recursion']        striver_a2z,neetcode_150
    4  trees              66   16  deep-study         ['recursion']        striver_a2z

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 RESULT: 2 adaptive mutations applied to a 4-task plan from live feeds.
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```

7. Viva points

Why a graph? Prerequisite chains and eligibility are naturally relational; inference computes what was not stored.

Why a fallback? External APIs have rate limits (GitHub allows 60 anonymous requests per hour per IP). The agent degrades instead of failing; that is an engineering requirement, not a defect.

The live values change between runs, so the numbers in this report are from one recorded run.

8. Conclusion

The agent combines reasoning over a knowledge graph with live external data, and its plan responds to both.